BCI project procedure - Preprocessing

P. Ofner, A. Schwarz, J. Pereira, GR. Müller-Putz (2017) ***Upper limb movements can be decoded from the time-domain of low-frequency EEG.*** *PLoS ONE 12(8): e0182578.*

EEG signals often contain artifacts that can compromise data quality. To address this, preprocessing transformations are typically applied. Traditional artifact removal relies on manual selection, a subjective and labor-intensive process. In contrast, automated pipelines ensure uniform artifact removal, improved efficiency in large datasets, and enhanced reproducibility across studies. Various software tools have been developed to automate different stages of EEG preprocessing: PREP, FASTER, ADJUST, ASR, MARA, SASICA.

The following workflow outlines a proposed EEG preprocessing pipeline, integrating established methods to ensure reliable and reproducible data processing.

DECIMATION →
- Reduce the sample frequency to a frequency close to 500Hz (not necessary in this case/project)

FILTERING →
- Electrical noise: **Notch** or adaptive filtering
- Band-pass filtering: 0.3 to 70 Hz

BAD CHANNEL REJECTION →
- What is a bad channel?
- Ofner 2017: Based on joint probability: threshold? outliers?
- Other approaches: correlation?

RE–REFERENCING →
- **Common Average Referencing** (not considering bad channels)
- Other approaches: Laplacian or REST

ICA →
- Reduction of contamination based on blind source separation
- Different ICA algorithms: Infomax vs **SOBI**
- Several origins: **Ocular** – muscular – channel noise – heart
- ICLabel probability: decide threshold (0.75, 0.8, 0.9)

INTERPOLATION OF BAD CHANNELS →
- Interpolation using the nearest electrodes
- Gold standard: **Spherical interpolation**

TRIAL EXTRACTION →
- Trials: time-locked segments synchronized with the presentation of the stimulus (stimulus-aligned trials) or the beginning of a movement (movement onset-aligned trials)
- Detection of movement onset
- Baseline correction

TRIAL REJECTION →
- Definition of bad trial: good trial if ok for all channels? Assess bad trials separately for each channel?
- Features: Amplitude? Kurtosis? Variance? Energy? Outliers?

DECIMATION

- Decimation is the process of reducing the sampling rate of a digital signal. This technique is commonly used in signal processing to decrease the amount of data that needs to be processed.
- Decimation involves lowpass-filtering a signal and then discarding some of its samples.
  The process typically includes downsampling, which refers to keeping only every Mth sample, where M is the decimation factor. To prevent aliasing, a lowpass filter is usually applied before downsampling. This filter removes frequency components above the new Nyquist frequency.You can use Matlab function "detrend".
- In Matlab, **decimate** and resample are both functions used to reduce the sampling rate of a signal, but they work differently. In general terms, if you need simple integer downsampling, decimate is efficient. If you need flexible resampling with upsampling/downsampling, resample is more versatile.
  - decimate (x, r): reduces the sampling rate of a signal x by a factor of r using a lowpass anti-aliasing filter before downsampling. It is mainly used for integer downsampling (e.g., reducing by a factor of 2, 4, etc.).
  - resample (x, p, q): changes the sampling rate of x by a rational factor p/q, meaning it can perform both downsampling and upsampling. It first interpolates the signal by p, then applies a lowpass filter, and finally downsamples by q.

FILTERING

- Notch filters are designed to attenuate or remove a specific narrow frequency band while minimizing the impact on adjacent frequencies. They are particularly useful for eliminating common electrical noise frequencies such as 50 Hz, 60 Hz, and their harmonics.
- **(OPTIONAL)** Adaptive notch filters improve upon traditional notch filters by automatically adjusting their parameters to optimize noise cancellation. Several studies have shown that adaptive notch filters outperform non-adaptive filters. You can consider adaptive filtering to reduce powerline contamination in this project. You have worked with adaptive filtering in the Biomedical Signals course.
- Bandpass filtering from 0.3Hz to 70Hz is crucial for several reasons:
  1) Noise elimination. This frequency range helps remove unwanted noise, improving data quality
  2) Elimination of DC components. The lower limit of 0.3 Hz helps eliminate low-frequency drifts and DC components, preserving saturation of the electronics;
  3) Preservation of relevant information. This frequency range retains most of the neurologically relevant information in the EEG signal, including delta, theta, alpha, beta, and part of the beta bands
  4) Improvement of signal-to-noise ratio. By eliminating frequencies outside the range of interest, the signal-to-noise ratio is improved, facilitating subsequent analysis of EEG data.
- In Matlab, you can use the **butter** function to designs bandpass and notch Butterworth filters, which are ideal for applications requiring a smooth frequency response without sharp

transtions. Once designed, the **filtfilt** function performs zero-phasing filtering by applying the filter twice (forward and backward), eliminating phase distortion and preserving the original signal shape while removing unwanted frequencies.

BAD CHANNEL REJECTION

- Bad channel rejection is a crucial step in EEG signal preprocessing to ensure data quality and accurate analysis. This step improves signal-to-noise ratio, prevents misinterpretation of experimental results, and enhances the accuracy of source reconstruction in EEG analysis.
- Methods for identifying bad channels:
  - Visual inspection: scrolling through channel data and examining channel spectra to identify flat or noisy channels.
  - Automated detection: Automated methods offer significant advantages over manual detection, including time-efficiency, consistency, and the ability to process large datasets. The vast majority of methods are based on computing different features for each channel and detecting and labeling as bad those channels with features that differ from the rest (outliers). Common features to identify bad channels include correlation coefficients, variance deviation, amplitude, and kurtosis. For example, Ofner et al., (2017) uses joint probability and the z-score (trimmed normalization) to identify the outliers (bad channels).

    Other approaches compute specific features related to bad channels and use machine learning algorithms for classification (Tuyisenge et al., 2018).

    Another alternative is an EEGLAB plugin which has been developed to detect and remove EEG bad channels using the Local Outlier Factor (LOF) algorithm (Kumaravel et al., 2022). Unlike global distribution methods, LOF identifies bad channels relative to local clusters, making it adaptable to various types of EEG data.
- We provide you with the list of bad channels for each subject and each run (see Atenea). We suggest using a methodology based on identifying outliers in one or multiple features (using z-score or Tukey's method) to label the bad channels.

  Identifying outliers in data is an important part of the analysis. An easy way to identify outliers in a data array is using the z-score for finding values which are a certain number of standard deviations away from the mean. The specific number of standard deviations is called the threshold (a recommended value is 5). In order to calculate robust values of mean and standard deviation (not distorted by outliers), you can use the trimmed normalization. Trimmed normalization is a method of averaging that removes a small designated percentage of the largest and smallest values before calculating the mean; i.e., we removed a predetermined amount of data values on each side of a distribution (a recommended value is 10%).

  Another outlier detection technique is the Tukey interquartile method, a simple rule of dumb based on the quartiles of the data. Tukey's rule says that the outliers are values more than 1.5 times the interquartile range (IQR: Q3-Q1) from the quartiles – either below Q1 – 1.5·IQR, or above Q3 + 1.5·IQR.

RE-REFERENCING

- Referencing of EEG data is a crucial preprocessing step that can significantly impact the quality and interpretation of EEG signals. Referencing is essentially a linear operation, irrespective of the reference electrode used during EEG acquisition, the EEG data can always be re-referenced to any other electrode (either physical or virtual) for further processing. Several re-referencing methods are commonly used:
  - Common Average Referencing (CAR): Computes the average of all electrode signals (not considering bad channels) as a reference. It reduces noise common to all channels. It improves signal-to-noise ratio, and it can be applied in real-time due to computational simplicity.
  - Mastoid reference: Often used as the original recording reference. It can be single or linked mastoids. It may introduce more noise than scalp channel references.
  - Double banana (longitudinal bipolar): Consists of multiple chains of bipolar channels running front to back. Useful for 10-20 electrode placement systems. It helps in localizing activity and reducing common noise.
  - Laplacian reference: Estimates the second derivative of scalp potential (five-point method or Hjorth technique, triangular method: using the three or four surrounding electrodes to estimate the average potential for each electrode, and the subtracting this average from the electrode's potential). It enhances spatial resolution and reduces volume conduction effects. It reduces broadband noise and improves signal-to-noise ratio. Sensitive to shallow local sources but less effective for deep sources.
  - Reference Electrode Standardization Technique (REST): Aims to approximate a point at infinity as a reference. It can resolve offline re-referencing issues in clinical EEGs, and it is particularly useful when the original reference is non-unipolar.
- Re-reference the EEG to CAR, optionally considering other re-referencing techniques.

ICA

- Independent Component Analysis (ICA) is a powerful technique for reducing nonneural contamination from EEG data. The key advantages of using ICA for artifact removal are:
  - It separates independent sources, including brain activity and artifacts.
  - It improves signal-to-noise ratio.
  - It can remove various types of artifacts, including ocular activity, muscle activity, heart, channel noise, and line noise.
  - It does not require a separate artifact reference signal.
  - It preserves more data compared to epoch rejection methodologies.
- ICLabel is an automated EEG independent component classifier which provides probability estimates for seven classes components: 1) brain, 2) eyes, 3) muscle, 4) heart, 5) line noise, 6) channel noise, and 7) other. These probabilities indicate the likelihood that a given independent component belongs to each class. The probabilities for each independent component sum to one across all seven categories. Use probability thresholds to identify components to remove., e.g., classifying an independent component as a blink artifact it its probability exceeds 0.7 for the eye category.

INTERPOLATION OF BAD CHANNELS

- Interpolation of bad channels is a critical preprocessing step to reconstruct signals from malfunctioning electrodes. Most common interpolation techniques and procedures are the following:
  - Neighbor Interpolation (NI): The concept is based on the similarity of EEG signals from adjacent electrodes. The procedure involves identifying 4-6 neighboring good channels around the bad channel, computing their average signal, and assigning this averaged signal to the bad channel. While this approach is quick and easy to implement, it is not highly accurate and may struggle when bad channels are scattered or adjacent to each other.
  - Spherical Spline Interpolation (SSI): The concept involves projecting electrode locations onto a sphere to estimate bad channel signals using spatial mapping. The procedure consists of three main steps: 1) mapping all electrode locations onto a unit sphere; 2) computing mapping matrices that relate good channels to bad ones; 3) using these matrices to estimate the signal for the bad channels. This method is widely used in EEG research due to its accuracy and its ability to handle multiple bad channels more effectively than NI. In terms of software implementation, it can be performed in EEGLAB through Tools > Interpolate electrodes.
- Use the spherical interpolation method to estimate the bad channels that were rejected at an earlier stage of the preprocessing pipeline.

TRIAL EXTRACTION

- Trial extraction in ERP experiments involves segmenting EEG data into time-locked trials based on specific events. Trials can be aligned with:
  - the presentation of a stimulus. Used when studying perceptual or cognitive processes.
  - the beginning of a response or of a movement.
- The complete trial extraction procedure can be summarized in the following points:
  - Event marking: The first step is to identify event timestamps in the continuous EEG data. Events can correspond to stimulus presentations or response or movements onsets, depending on the experiment's design.
  - Epoching: Once events are marked, time windows (epochs or trials) are extracted around each event. These epochs include both a pre-stimulus (or pre-movement) period and a post-stimulus period, ensuring a comprehensive analysis of event-related responses.
  - Detection of movement onset: In movement-related studies, detecting the precise onset of movement is essential. Several methods can be used for this purpose, including:
    - EMG threshold crossing, which identifies muscle activation onset.
    - Accelerometer data analysis, which tracks motion patterns.
    - Video-based motion tracking, which visually detects motion initiation.
  - Baseline correction: To improve signal stability and comparability across trials, baseline correction is applied. This involves subtracting the mean of a pre-stimulus interval from each trial. Common baseline periods include -200 to 0 ms or -100 to 0 ms relative to the event onset. This step helps to reduce slow drifts and normalize the data for better interpretation.

- Signal segmentation: Form the trials using the stimulus sample (2nd column of the field "events" in the MAT file) and the event code of the class movement. The duration of the trials must be from -2.5 to 6 seconds, considering 0 the time sample where the stimulus related to the desired movement was presented. If you work with a 3D matrix (recommended), you should have a matrix with 60 trials x 61 channels x 4352 samples.

TRIAL REJECTION

- This process involves removing trials contaminated by artifacts or errors to improve signal quality and ensure accurate results. Key aspects of trial rejection include:
  - Artifact rejection and removal
    - Threshold-based: Trials exceeding predefined voltage amplitude thresholds are marked for rejection.
    - Statistical-based: Using features (kurtosis, energy or joint probability) to identify outlier trials. Rejecting trials with extreme values is effective at reducing noise.
  - Erroneous response removal: trials with incorrect behavioral responses should be removed.
  - Component-specific considerations: different ERP components (e.g., P300, N400, N170, MRCP, ERD/ERS) may require tailored rejection criteria.
  - Automated and manual approaches: combining automated rejection methods with manual review for optimal results.
- The purpose of this step is to identify the free-of-artifact trials for each channel, class movement and subject. An easy way to work is to generate a 2D matrix (60 trials x 61 channels) containing 0 or 1 (e.g., 0 artifact and 1 non-artifact). In other words, if trial $\boldsymbol{i}$ is marked as bad for the channel $\boldsymbol{j}$, then the element ($\boldsymbol{i}$, $\boldsymbol{j}$) should be 0.
- Mark trials as bad if:
  - The physical movement starts before 100ms from the appearance of the stimulus (considered the minimum reaction time). We considered that the subject comes early if reaction time is lower than 100ms.
  - The movement starts after 2s from the appearance of the stimulus. We considered that subject does not process this stimulus.
- To identify bad trials for each channel, you can use either **threshold-based techniques** (e.g., ensuring that the trial amplitude does not exceed ±150μV) or **outlier detection methods** based on different features computed from the trials, such as amplitude, variance, or kurtosis.

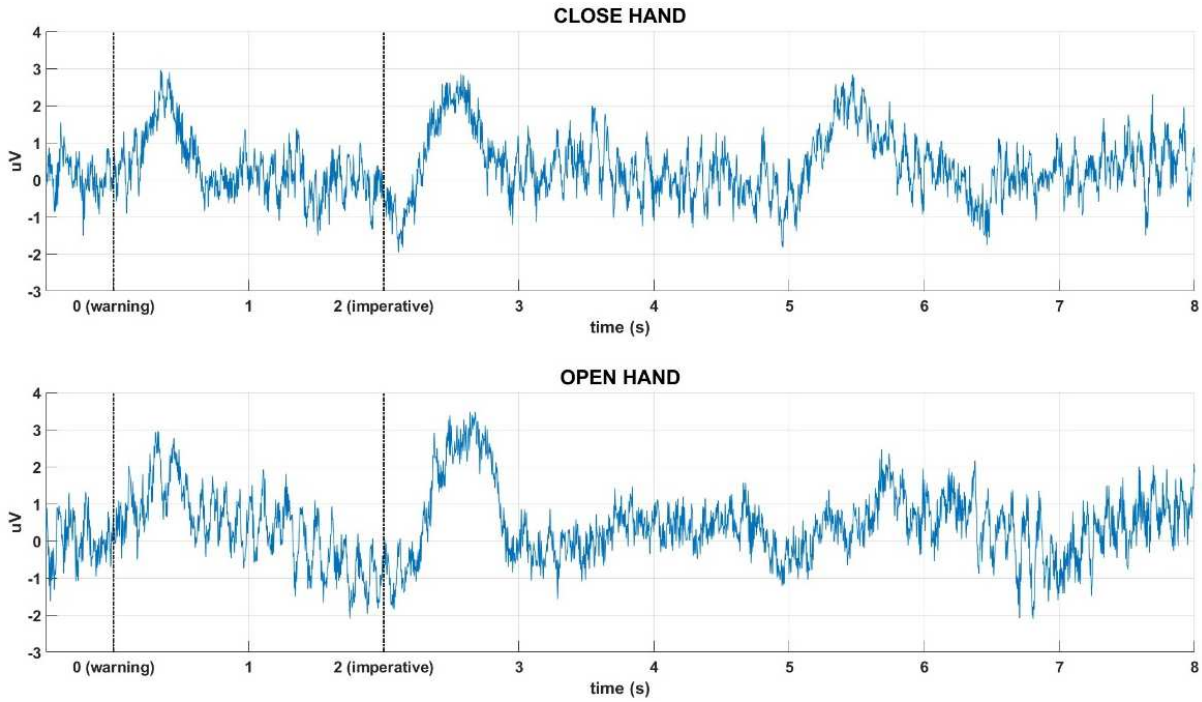
  Here are some examples of good trial percentages for comparison with your results. However, keep in mind that these values may vary depending on some preprocessing steps (filtering, ICA, interpolation, etc.). The table below presents the percentage of artifact-free trials for channels FC4 (17), FCC2h (23), Cz (30), and CCP6h (40) during hand movements.

SUBJECT	CLOSE HAND (1540)				OPEN HAND (1541)			
	FC4	FCC2h	Cz	CCP6h	FC4	FCC2h	Cz	CCP6h
S1	96.67%	88.33%	96.67%	93.33%	98.83%	93.33%	100%	90.00%
S3	83.33%	90.00%	98.33%	95.00%	88.33%	90.00%	98.33%	98.33%
S4	91.67%	96.67%	96.67%	88.33%	96.67%	98.33%	100%	93.33%

DELIVERABLES

D1. Artifacted channels for each subject and run

D2. Percentage of good (artifact-free) trials for each subject and movement class

D3. Plot the average (considering only the good trials) for channel Cz (30), for each subject and class movement. Plot also the grand-mean average (average of all subjects) of trials for Cz and each movement class.

The figure below presents an example of the stimulus-locked average ERP at channel Cz for hand movements of subject S1.